

Pack the Prism



Grade 5 • Mathematics
NC.5.MD.5

Connect layers of cubes with volume.



What will a strong explanation show?

1. A model or equation that fits the quantities.
2. Accurate calculation with appropriate units.
3. A check that makes the answer reasonable.



Watch the reasoning

A storage prism for the library team measures 4 cm by 3 cm by 3 cm. Find its volume. Explain the number of unit cubes in one layer and the number of layers.

One layer has $4 \times 3 = 12$ cubes. 3 layers give $12 \times 3 = 36 \text{ cm}^3$.



Build a second model together

A storage prism for the art club measures 7 cm by 4 cm by 2 cm. Find its volume. Explain the number of unit cubes in one layer and the number of layers.

Discuss: What should the first step find?



Your turn • 1

A storage prism for the art club measures 6 cm by 3 cm by 7 cm. Find its volume. Explain the number of unit cubes in one layer and the number of layers.

Represent → solve → explain → check



Your turn • 2

A storage prism for the science club measures 3 cm by 8 cm by 2 cm. Find its volume. Explain the number of unit cubes in one layer and the number of layers.

Represent → solve → explain → check



Your turn • 3

A storage prism for the community center measures 6 cm by 5 cm by 6 cm. Find its volume. Explain the number of unit cubes in one layer and the number of layers.

Represent → solve → explain → check



Your turn • 4

A storage prism for the reading group measures 7 cm by 6 cm by 7 cm. Find its volume. Explain the number of unit cubes in one layer and the number of layers.

Represent → solve → explain → check



Exit question • Explain independently

A storage prism for the music club measures 5 cm by 4 cm by 6 cm. Find its volume. Explain the number of unit cubes in one layer and the number of layers.

Include a check. What would you teach someone else?



Teacher review and next steps

1. 126 cm³

2. 48 cm³

3. 180 cm³

4. 294 cm³

Exit: 120 cm³

Use the worked PDF key and slide notes to discuss methods. Revisit the earliest unclear step before adding more practice.

